NewMessage changeset, 128, 130, 172–174
nil data type, 15
Nx (Numerical Elixir), 190

O

Oban library, 66, 81, 159–169, 183–185
object-oriented language, Ruby as, 14, 41
objects vs. GenServer, 79
observability, 167, 188
one-to-many Ecto schema association, 96
one-to-one Ecto schema association, 96
Open Telecom Platform (OTP), 8, 10
operators, 16–18
-otp suffix, 10
overloading functions, 22–25, 60–64

P

~p sigil, 127
PageController route, 126
parallel and serial code execution, 7
parallelism, *see also* GenServers
  being cautious with, 81
  and concurrency, 65–70
  Elixir's parallelism model, 69
  Ruby's parallelism model, 68
pattern matching, 49–64
  control flow, 55–60
  function overloading, 60–64
  lists and tuples, 52–53
  map matching, 53
  and operators, 16, 50–52
  pinned values, 54
persisting data, *see* Ecto library
PgBouncer library, 142
Phoenix library, 117–140
  authentication generator, 138
  Channels, 119, 139
  components for a clean UI, 134–138
  deployment guidance, 188
  foundations of, 117–121
  generators, 93
  and GenServers, 66
  LiveView, 119, 139, 189
  vs. Rails, 119, 123
  routing requests, 122–125
  serving requests with controllers, 125–132
  SMS sample app build, 89
  static asset management, 132–134
  testing an Ecto query, 175–176
  testing requests, 179–183
Phoenix.Component module, 134
Phoenix.Endpoint module, 122
Phoenix.HTML, 118
Phoenix/Ecto package, 119
phoenix_live_view package, 118
PhoneApp, 122
PhoneApp.Conversations context, 113
PhoneApp.Repo module, 105
PhoneAppWeb, 122
PhoneAppWeb.CoreComponents module, 138
PhoneAppWeb.Endpoint module, 122
PhoneAppWeb.MessageHTML module, 128
PhoneAppWeb.Router module, 123
pinned values, 54
pipe (|) operator, 17, 52
pipeline function, 124
plug function, 123, 126
Plug library, 118, 122
Plug.Conn struct, 127
plugins, Oban, 162
pooled connections, HTTP requests, 142
port for route requests, 121
POST requests, testing, 182
Postgres, and Oban, 159–160, 166
pretrained neural network models, 191
Process.info(pid), 74
processes in Elixir, 70–76
  concurrency and parallelism, 69–70
  error isolation, 73
  garbage collection, 74–76
  GenServer library, 76–80
  memory architecture, 74
  recursion and processing messages forever, 72
  spawning, 70–72
Procs vs. anonymous functions, 22
protocols, 43, 142

Q

queries, Ecto, 91–92, 105–112, 175–176
queues, Oban, 162

R

race conditions, Ruby's concurrency model, 67
Rack library, 118
Ractors, Ruby, 69
Rails (Ruby on Rails), 118–119, 123, 125–126
raise operator, 17
Rake tool, Ruby, 19
real-time web applications, 119
receive function, 71
recovering stuck async jobs, 168
recursion, 63–64, 72
reductions, in BEAM parallel processing, 70
render function, 129
repeating tasks, optimizing with async job, 155
Repo module, 105
Repository design pattern, 91
Req HTTP request client, *see* HTTP requests
requests, serving, *see* Phoenix library
require module keyword, 28
required keys, structs, 40
return syntax, Ruby vs. Elixir, 23
routing requests, 122–125
RSpec, 172
Ruby, 3–5
  ActiveRecord, 87, 91–92, 99, 110
  concurrency in, 65, 67
  data types, 14
  enumeration syntax, 45